

NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Third Semester B.E. (Civil) (Credit System) Degree Examinations

Supplementary Examinations - December 2025

21CV304 / 20CV303 - STRENGTH OF MATERIALS

Duration: 3 Hours

Max. Marks: 100

*Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.*

Unit – I

1. a) Define the terms: Poisson's ratio and Modulus of elasticity.
- b) A circular rod of diameter 100 mm is 1.0 m long and is subjected to a compressive load of 50 kN. Determine the stress and strain if $E = 205 \text{ GPa}$.
- c) Different portions of a stepped bar are subjected to forces as shown in Figure 1 (c). Determine stresses induced in each portion and net deformation of the bar if the modulus of elasticity of the material is $2 \times 10^5 \text{ N/mm}^2$.

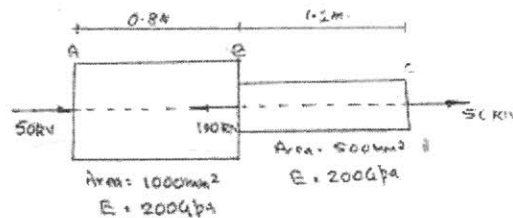


Figure 1 (c)

2. a) Draw the typical stress strain diagram for mild steel and show salient points.
- b) For a given material, modulus of elasticity is 210 GPa and shear modulus is 80 GPa. Find the Bulk modulus and reduction in diameter of round bar of diameter 50 mm and 1200 mm long when stretched by 1.3 mm.
- c) Calculate normal and Shear stress on the plane AB as shown in Figure 3 (c).

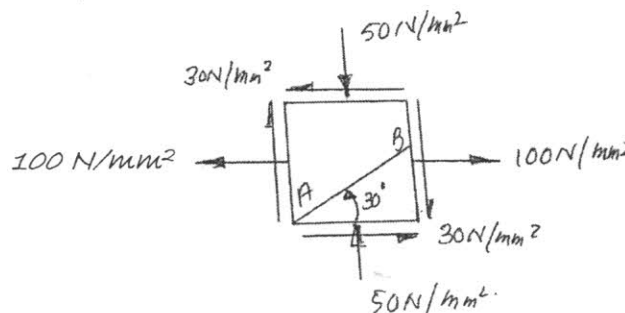


Figure 3 (c)

3. a) State any four assumptions that are made in Lamé's theory of thick cylinders.
- b) Define principal stresses and planes. Give their expressions in two-dimensional stress system.
- c) A thin cylinder of 300 mm inside diameter is subjected to an internal pressure of 6 N/mm². The permissible stress in the material is 50 N/mm². Determine the wall thickness.

P.T.O.

Marks	BT*	CO*	PO*
04	L*2	1	1
06	L3	1	3

10	L3	2	2
04	L2	1	2
06	L3	1	3

10	L3	02	02
04	L2	2	1
06	L2	2	1
10	L3	2	2

Unit – II

- 4 a) A beam AB is loaded as shown in Figure 4 (b). Draw SFD and BMD showing salient features.

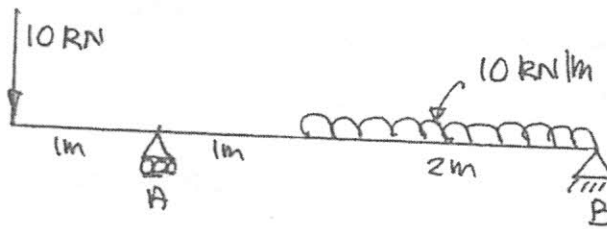


Figure 4 (a)

- b) A cantilever beam of length L is subjected to udl W/m for the entire span. Determine the maximum deflection.

10 L3 3 2

10 L3 4 2

5. a) A beam AB is loaded as shown in Figure 5 (b). Draw SFD and BMD showing salient features.

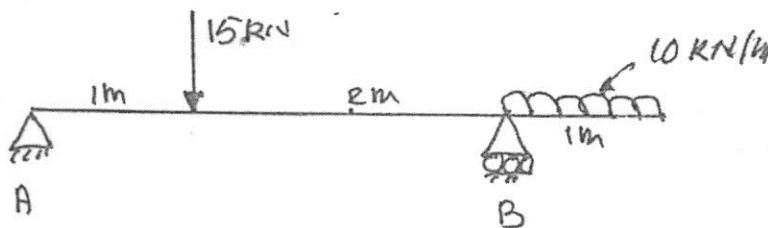


Figure 5 (a)

- b) A simply supported beam has a rectangular cross section 120 mm x 200 mm. Determine the maximum bending moment that can be imposed on the section if the compressive stress is not to exceed 60 N/mm². Draw the bending stress distribution across the section.

10 L3 3 2

10 L3 4 3

6. a) A simply supported beam of length L is subjected to udl W/m for the entire span. Determine the maximum slope and deflection.

10 L3 3 2

- b) A simply supported beam of length 4 m is simply supported at its ends and carries a point load of 30 kN at mid span. Find the deflection at mid span and slope at supports. Take $E = 200 \text{ GPa}$ and $I = 10 \times 10^7 \text{ mm}^4$.

10 L3 4 3

Unit – III

7. a) A hollow circular shaft with diameter ratio 0.7 has to transmit a power of 200 kW at 110 rpm. Find the diameter of the shaft, if the shear stress of the material must not exceed 50 N/mm² and the angle of twist 1 degree in a length of 3.0 m. Take $G = 90 \text{ GPa}$.

10 L2 5 1

- b) Compare the loads given by Euler's formula and Rankine's formula for a column, 3m long having inner diameter 40 mm and outer diameter 60 mm. The column is fixed at one end and hinged at the other. Take $\sigma_c = 350 \text{ N/mm}^2$ and $\alpha = 1/7500$ and $E = 200 \text{ GN/m}^2$.

10 L3 5 3

8. a) Show that hollow shaft is stronger than the solid shaft in transmitting torque, having same length and weight, made of same material.

10 L2 5 2

- b) A column, 3m long having inner diameter 50 mm and outer diameter 70 mm. Calculate the safe load a column can carry with a factor of safety 3 using Euler's equation for the following end conditions. Assume $E = 200 \text{ GPa}$.

a) Both ends fixed

b) Both ends hinged

c) One end hinged and another end free.

10 L3 5 3